
GENERAL RELATIVITY

Tutorial problem set 6, 16.10.2020, with solutions

■ PROBLEM 16 Paraboloidal coordinates.

In Euclidean 3-dimensional space, we can define the so-called *paraboloidal coordinates* (u, v, ϕ) by

$$x = uv \cos \phi, \quad y = uv \sin \phi, \quad z = \frac{1}{2}(u^2 - v^2), \quad (16.1)$$

where the ranges of coordinates are $u \in [0, \infty)$, $v \in [0, \infty)$, $\phi \in [0, 2\pi)$.

- (a) Find the coordinate transformation matrix between paraboloidal and Cartesian coordinates $\partial x^\alpha / \partial \tilde{x}^\beta$ and the inverse transformation. Are there any singular points in the map?
- (b) Find the basis vectors in paraboloidal coordinates in terms of Cartesian basis vectors.
- (c) Find the metric and inverse metric in paraboloidal coordinates.
- (d) Calculate the Christoffel symbols in paraboloidal coordinates.
- (e) Express the divergence $\nabla_\mu V^\mu$ and Laplacian $\nabla_\mu \nabla^\mu f$ in paraboloidal coordinates.

SOLUTION: (a) Differentiating Eqs. (16.1) one obtains,

$$\begin{aligned} dx &= v \cos(\phi) du + u \cos(\phi) dv - uv \sin(\phi) d\phi \\ dy &= v \sin(\phi) du + u \sin(\phi) dv + uv \cos(\phi) d\phi, \\ dz &= u du - v dv, \end{aligned} \quad (16.2)$$

from which one obtains the matrix that transforms the parabolic coordinates into the Cartesian coordinates,

$$\begin{pmatrix} v \cos(\phi) & u \cos(\phi) & -uv \sin(\phi) \\ v \sin(\phi) & u \sin(\phi) & uv \cos(\phi) \\ u & -v & 0 \end{pmatrix}, \quad (16.3)$$

The determinant of this transformation is $uv(u^2 + v^2)$, which means that $u = 0$ (with arbitrary v and ϕ) and $v = 0$ (with arbitrary u and ϕ) are the (sets of) singular points of the transformation. The inverse transformation can be obtained by taking a differential of,

$$\tan(\phi) = \frac{y}{x}, \quad r \equiv \sqrt{x^2 + y^2 + z^2} = \frac{1}{2}(u^2 + v^2) \implies u^2 = r + z, \quad v^2 = r - z. \quad (16.4)$$

With $dr = \frac{x}{r} dx + \frac{y}{r} dy + \frac{z}{r} dz$ one can take a differential of (16.4) to get the inverse transformation,

$$\begin{pmatrix} \frac{x}{2ur} & \frac{y}{2ur} & \frac{z+r}{2ur} \\ \frac{x}{2ur} & \frac{y}{2ur} & \frac{z-r}{2ur} \\ -\frac{y}{x^2+y^2} & \frac{x}{x^2+y^2} & 0 \end{pmatrix} = \begin{pmatrix} \frac{v \cos(\phi)}{u^2+v^2} & \frac{v \sin(\phi)}{u^2+v^2} & \frac{u}{(u^2+v^2)} \\ \frac{u \cos(\phi)}{u^2+v^2} & \frac{u \sin(\phi)}{u^2+v^2} & -\frac{v}{(u^2+v^2)} \\ -\frac{\sin(\phi)}{uv} & \frac{\cos(\phi)}{uv} & 0 \end{pmatrix}, \quad (16.5)$$

(b) We need to find the matrix corresponding to the transformation, $\tilde{\partial}_\mu = \frac{\partial x^\alpha}{\partial \tilde{x}^\mu} \partial_\alpha$, where $(\tilde{\partial}_\mu) = (\partial_u, \partial_v, \partial_\phi)$ and $(\partial_\alpha) = (\partial_x, \partial_y, \partial_z)$, which is the same transformation matrix that transforms $d\tilde{x}^\mu$ into dx^α , *i.e.* $dx^\alpha = \frac{\partial x^\alpha}{\partial \tilde{x}^\mu} d\tilde{x}^\mu$. Hence, this matrix is given in Eq. (16.3).

(c) This is a flat metric with the line element, $ds^2 = dx^2 + dy^2 + dz^2$. From Eq. (16.2) it then follows,

$$ds^2 = (u^2 + v^2) [du^2 + dv^2] + (uv)^2 d\phi^2, \quad (16.6)$$

The inverse metric is given by,

$$g^{\mu\nu} = \text{diag} \left(\frac{1}{u^2 + v^2}, \frac{1}{u^2 + v^2}, \frac{1}{(uv)^2} \right). \quad (16.7)$$

(d) The non-vanishing Christoffel symbols associated with the line element (16.6) are,

$$\begin{aligned} \Gamma_{uu}^u &= \frac{u}{(u^2 + v^2)}, & \Gamma_{uv}^u &= \Gamma_{vu}^u = \frac{v}{(u^2 + v^2)} \\ \Gamma_{vv}^u &= -\frac{u}{(u^2 + v^2)}, & \Gamma_{uu}^v &= -\frac{v}{(u^2 + v^2)} \\ \Gamma_{uv}^v &= \Gamma_{vu}^v = \frac{u}{(u^2 + v^2)}, & \Gamma_{vv}^v &= \frac{v}{(u^2 + v^2)} \\ \Gamma_{\phi\phi}^u &= -\frac{uv^2}{(u^2 + v^2)}, & \Gamma_{\phi\phi}^v &= -\frac{u^2v}{(u^2 + v^2)} \\ \Gamma_{u\phi}^\phi &= \Gamma_{\phi u}^\phi = \frac{1}{u}, & \Gamma_{v\phi}^\phi &= \Gamma_{\phi v}^\phi = \frac{1}{v} \end{aligned} \quad (16.8)$$

(e) Notice first that, $\nabla_\mu V^\mu = \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} V^\mu)$ and $\nabla_\mu \nabla^\mu f = \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} g^{\mu\nu} \partial_\nu f)$, where $\sqrt{g} = (u^2 + v^2)uv$. With these in mind we then have,

$$\nabla_\mu V^\mu = \left[\partial_u V^u + \partial_v V^v + \frac{(3u^2 + v^2)}{(u^2 + v^2)u} V^u + \frac{(u^2 + 3v^2)}{(u^2 + v^2)v} V^v \right] + \partial_\phi V^\phi \quad (16.9)$$

and

$$\nabla_\mu \nabla^\mu f = \frac{1}{u^2 + v^2} \left[\partial_u^2 + \partial_v^2 + \frac{1}{u} \partial_u + \frac{1}{v} \partial_v \right] f + \frac{1}{(uv)^2} \partial_\phi^2 f. \quad (16.10)$$

■ PROBLEM 17 Geodesics in Rindler space-time.

Let (x^0, x) denote Minkowski coordinates in an inertial system **I**, and let (w^0, w) denote the Rindler coordinates of a system of reference **R** with constant acceleration g relative to **I**. The two systems are related by

$$x = \frac{c^2}{g} \left[\cosh \left(\frac{gw^0}{c^2} \right) - 1 \right] + w \cosh \left(\frac{gw^0}{c^2} \right), \quad (17.1)$$

$$x^0 = \frac{c^2}{g} \sinh \left(\frac{gw^0}{c^2} \right) + w \sinh \left(\frac{gw^0}{c^2} \right). \quad (17.2)$$

You have derived in Homework 1, Problem 1, that the invariant element in Rindler coordinates is,

$$ds^2 = - \left(1 + \frac{gw}{c^2} \right)^2 (dw^0)^2 + dw^2 \quad (17.3)$$

- (a) Write down the equation of motion for a particle in free fall as a second order differential equation in w^0 .
- (b) Show that the solution for a particle starting at $w = \bar{w}$ at time $w^0 = 0$ with velocity zero is

$$w(w^0) = \frac{c^2}{g} \left[\left(1 + \frac{g\bar{w}}{c^2}\right) \frac{1}{\cosh\left(\frac{gw^0}{c^2}\right)} - 1 \right]. \quad (17.4)$$

- (c) Calculate the velocity of the particle as a function of w^0 ($v = cdw/dw^0$). Find the maximum velocity of the particle and compare with the velocity of light. What happens for $w^0 \rightarrow \infty$.
- (d) Calculate the proper time of the particle as a function of w^0 and $\bar{w}$.

SOLUTION: (a) Upon differentiating Eqs. (17.1–17.2) one obtains,

$$dx = \left(1 + \frac{gw}{c^2}\right) \sinh\left(\frac{gw^0}{c^2}\right) dw^0 + \cosh\left(\frac{gw^0}{c^2}\right) dw \quad (17.5)$$

$$dx^0 = \left(1 + \frac{gw}{c^2}\right) \cosh\left(\frac{gw^0}{c^2}\right) dw^0 + \sinh\left(\frac{gw^0}{c^2}\right) dw \quad (17.6)$$

A free falling particle is moving with a constant velocity $dx/dx^0 = v(0)/c = \text{const}$. This can be used in (17.5–17.6) to obtain a differential equation for dw/dw^0 in the form,

$$\frac{dw}{dw^0} = \left(1 + \frac{gw}{c^2}\right) \frac{\frac{v(0)}{c} - \tanh\left(\frac{gw^0}{c^2}\right)}{1 - \frac{v(0)}{c} \tanh\left(\frac{gw^0}{c^2}\right)}. \quad (17.7)$$

Differentiating this once gives the desired general equation for $d^2w/(dw^0)^2$, which is complicated. While Eq. (17.7) holds for general trajectories, the problem at hand asks for simpler trajectories, namely for those for which the initial velocity vanishes, $dw(0)/dw^0 = 0$. From Eq. (17.7) we see that that means that $v(0) = 0$ and thus,

$$\frac{dw}{dw^0} = - \left(1 + \frac{gw}{c^2}\right) \tanh\left(\frac{gw^0}{c^2}\right). \quad (17.8)$$

Differentiating this once gives the asked-for differential equation,

$$\begin{aligned} \frac{d^2w}{(dw^0)^2} &= -\frac{g}{c^2} \left[\left(1 + \frac{gw}{c^2}\right) \frac{1}{\cosh^2\left(\frac{gw^0}{c^2}\right)} + \frac{dw}{dw^0} \tanh\left(\frac{gw^0}{c^2}\right) \right] \\ &= -\frac{g}{c^2} \left(1 + \frac{gw}{c^2}\right) \left[\frac{2}{\cosh^2\left(\frac{gw^0}{c^2}\right)} - 1 \right]. \end{aligned} \quad (17.9)$$

Since this problem is of profound importance for understanding the physics of black holes, we shall present an alternative derivation, which makes use of the geodesic equations. The non-vanishing Christoffel symbols are,

$$\Gamma_{01}^0 = \Gamma_{10}^0 = \frac{g/c^2}{1 + gw/c^2}, \quad \Gamma_{00}^1 = -\frac{g/c^2}{1 + gw/c^2} \quad (17.10)$$

such that the geodesic equations are,

$$\frac{d^2 w^0}{d\tau^2} + \frac{2g/c^2}{1 + gw/c^2} \frac{dw^0}{d\tau} \frac{dw}{d\tau} = 0 \quad (17.11)$$

$$\frac{d^2 w}{d\tau^2} - \frac{g/c^2}{1 + gw/c^2} \frac{dw^0}{d\tau} \frac{dw^0}{d\tau} = 0 \quad (17.12)$$

Notice that, upon multiplying the first geodesic equation (17.11) by $(1 + gw/c^2)^2$, one can rewrite it as,

$$\frac{d}{d\tau} \left[\left(1 + \frac{gw}{c^2}\right)^2 \frac{dw^0}{d\tau} \right] = 0, \quad (17.13)$$

which means that the quantity,

$$\left(1 + \frac{gw}{c^2}\right)^2 \frac{dw^0}{d\tau} = \frac{p_0}{m} = \text{const.} \quad (17.14)$$

is conserved. The physical interpretation is that the 0-th component of the momentum of a particle (written as a 1-form with the index down) is conserved. This can be reinterpreted as a conserved energy. The origin of this conserved quantity can be understood as follows. In space-times in which the metric is time independent (in the context of field theory, the metric tensor can be viewed as a background field on which the Hamiltonian depends on a certain well-defined way), $p_0 \equiv E/c$ is conserved. Similarly, if the metric does not depend on a coordinate x^i , then p_i will be conserved. That also means that there is a global symmetry associated with the metric, corresponding to translations along the direction x^i . These global symmetries of the metric tensor are known as isometries, and their generators are known as Killing vectors. Later in the course we will learn to construct all of the isometries of a metric.

To these considerations, one can add the line element (17.3), which can be thought of as the integral of motion of the geodesic equations, in the form

$$c^2 = \left(1 + \frac{gw}{c^2}\right)^2 \left(\frac{dw^0}{d\tau}\right)^2 - \left(\frac{dw}{d\tau}\right)^2. \quad (17.15)$$

where $d\tau^2 = -ds^2/c^2$ is the proper time. Now taking account of the conserved energy E (which for notational convenience we define the energy per unit mass),

$$\frac{E}{c} = \left(1 + \frac{gw}{c^2}\right)^2 \frac{dw^0}{d\tau}. \quad (17.16)$$

and inserting this into Eq. (17.15) one obtains,

$$c^2 = \frac{(E/c)^2}{\left(1 + \frac{gw}{c^2}\right)^2} - \left(\frac{dw}{d\tau}\right)^2 = \frac{(E/c)^2}{\left(1 + \frac{gw}{c^2}\right)^2} - \frac{(E/c)^2}{\left(1 + \frac{gw}{c^2}\right)^4} \left(\frac{dw}{dw^0}\right)^2 \quad (17.17)$$

which is a first order differential equation for dw/dw^0 ,

$$\left(\frac{dw}{dw^0}\right)^2 = \left(1 + \frac{gw}{c^2}\right)^2 \left[1 - \frac{c^4}{E^2} \left(1 + \frac{gw}{c^2}\right)^2\right]. \quad (17.18)$$

Differentiating this once, gives the desired second order equation,

$$\frac{d^2 w}{(dw^0)^2} = \frac{g}{c^2} \left(1 + \frac{gw}{c^2}\right) \left[1 - \frac{2c^4}{E^2} \left(1 + \frac{gw}{c^2}\right)^2\right]. \quad (17.19)$$

When $(dw/dw^0)_{w^0=0} = 0$ is inserted, this equation should be equivalent with Eq. (17.9). We leave as an exercise to show that the two equations are indeed equivalent.

(b) To accomplish the task, we need to integrate Eq. (17.18), which can be conveniently rewritten as,

$$\int \frac{dw}{\left(1 + \frac{gw}{c^2}\right) \sqrt{1 - \frac{c^2}{E^2} \left(1 + \frac{gw}{c^2}\right)^2}} = \pm \int dw^0. \quad (17.20)$$

This integral simplifies to $\int dy/[y\sqrt{1-y^2}]$ for the variable, $y = \frac{c}{E} \left(1 + \frac{gw}{c^2}\right)$. Its primitive function is, $-\text{arctanh}(\sqrt{1-y^2})$ such that we have,

$$\sqrt{1 - \frac{c^4}{E^2} \left(1 + \frac{gw}{c^2}\right)^2} = \mp \tanh\left(\frac{gw^0}{c^2}\right) + \text{const.} = \mp \sqrt{1 - \frac{1}{\cosh^2(gw^0/c^2)}} + \text{const.} \quad (17.21)$$

If the initial position is $w(0) = \bar{w}$, then $(\text{const.})^2 = 1 - \frac{c^4}{E^2} (1 + g\bar{w}/c^2)^2$ and the general solution becomes rather complicated.

The solution simplifies considerably if the initial velocity vanishes, $(dw/dw^0)_{w^0=0} = 0$. which corresponds to the problem at hand. (When viewed on the space-time diagram, the set of all solutions are vertical geodesics which cross the real axis at some $w(0) = \bar{w}$). In this case, $\text{const.} = 0$, $(E/c^2) = 1 + \frac{g\bar{w}}{c^2}$ and we have,

$$1 + \frac{gw}{c^2} = \frac{E}{c^2} \frac{1}{\cosh\left(\frac{gw^0}{c^2}\right)} \implies w(w^0) = \frac{c^2}{g} \left[\frac{E}{c^2} \frac{1}{\cosh(gw^0/c^2)} - 1 \right]. \quad (17.22)$$

which can be also written as,

$$w(w^0) = \frac{c^2}{g} \left[\frac{1 + \frac{g\bar{w}}{c^2}}{\cosh\left(\frac{gw^0}{c^2}\right)} - 1 \right], \quad (17.23)$$

which agrees with Eq. (17.4).

(c) The particle's velocity is given by,

$$v = c \frac{dw}{dw^0} = -c \left(1 + \frac{g\bar{w}}{c^2}\right) \frac{\sinh(gw^0/c^2)}{\cosh^2(gw^0/c^2)} \quad (17.24)$$

As $w^0 \rightarrow \infty$, we see that $v \rightarrow 0$, *i.e.* the coordinate velocity vanishes. This is analogous to what happens at the event horizon of a black hole, where the coordinate velocity of a particle (when written with respect to the time of asymptotic observer) vanishes, because of an infinite time redshift that the coordinate time (of an observer far away from the black hole) exhibits.

From (17.24) one can see that the maximum velocity is reached when,

$$\cosh\left(\frac{gw^0}{c^2}\right) = \sqrt{2}, \quad \sinh\left(\frac{gw^0}{c^2}\right) = \pm 1. \quad (17.25)$$

Inserting this into (17.24) gives

$$v_{\max} = \mp \frac{c}{2} \left(1 + \frac{g\bar{w}}{c^2}\right), \quad (17.26)$$

which can be (in magnitude) larger or smaller than c depending on the choice of $\bar{w}$. This should serve as a warning when trying to give a physical interpretation of a coordinate velocity. Note that the coordinates in the right Rindler wedge are limited to $\bar{w} \geq -c^2/g$.

(d) Equation (17.17) can be recast as,

$$\left(\frac{dw}{d\tau}\right)^2 = \frac{(E/c)^2}{\left(1+\frac{gw}{c^2}\right)^2} - c^2 \implies \frac{\left(1+\frac{gw}{c^2}\right) dw}{\sqrt{\frac{E^2}{c^4} - \left(1+\frac{gw}{c^2}\right)^2}} = \pm c d\tau \quad (17.27)$$

Integrating this yields,

$$\frac{E^2}{c^4} = \left(1+\frac{g\bar{w}}{c^2}\right)^2 = \left(1+\frac{gw}{c^2}\right)^2 + \left(\frac{g\tau}{c}\right)^2, \quad (17.28)$$

where we took account of the above initial conditions. From this we see that the proper time coordinates are limited to,

$$-\frac{c}{g} \left(1+\frac{g\bar{w}}{c^2}\right) < \tau < \frac{c}{g} \left(1+\frac{g\bar{w}}{c^2}\right), \quad (17.29)$$

which seems puzzling.

From the Rindler space line element (17.3) we have,

$$\left(\frac{d\tau}{dw^0}\right)^2 = \frac{1}{c^2} \left[\left(1+\frac{gw}{c^2}\right)^2 - \left(\frac{dw}{dw^0}\right)^2 \right] \quad (17.30)$$

Inserting into this the result (17.23) we found in part (b), one obtains,

$$d\tau = \pm \frac{1}{c} \left(1+\frac{g\bar{w}}{c^2}\right) \frac{dw^0}{\cosh^2\left(\frac{gw^0}{c^2}\right)}. \quad (17.31)$$

Integrating this yields,

$$\frac{g\tau}{c} = \pm \left(1+\frac{g\bar{w}}{c^2}\right) \tanh\left(\frac{gw^0}{c^2}\right). \quad (17.32)$$

From this we see that the proper time coordinates,

$$-\frac{c}{g} \left(1+\frac{g\bar{w}}{c^2}\right) < \tau < \frac{c}{g} \left(1+\frac{g\bar{w}}{c^2}\right) \quad (17.33)$$

cover the whole range $-\infty < w^0 < \infty$ of Rindler coordinates, thus explaining thus the coordinate range puzzle (17.29). In fact, it is easy to understand this result when one draws the world lines on these geodesics, which appear as vertical lines on Minkowski space-time. If one adds the right Rindler wedge and the curves of constant w and w^0 on it, one will realize that curves reach the upper (lower) Rindler at $w^0 \rightarrow +\infty$ ($w^0 \rightarrow -\infty$), where the geodesic stops. Hence, the upper and lower Rindler wedge act as the future and past Cauchy horizon of Rindler space. However, these are the horizons associated with Rindler coordinates. Indeed, Rindler coordinates can be smoothly extended. A simple smooth extension of Rindler coordinates, are Cartesian coordinates of Minkowski space, in terms of which vertical geodesics can be written as $x(\tau) = x(0)$. These are exactly the same geodesics we found above in Eq. (17.23), but now rewritten in Cartesian coordinates. Obviously, nothing strange happens at the Rindler wedges,

$$\tau_{\text{RW}\pm}(\bar{w}) = \pm \frac{c}{g} \left(1+\frac{g\bar{w}}{c^2}\right) \quad (17.34)$$

and particles will just happily pass through them. As we will see, an analogous phenomenon occurs in black hole metrics: static (*e.g.* Schwarzschild) coordinates become singular at the event horizon, but that is a consequence of an infinite redshift that the coordinate time of an asymptotic observer suffers at the event horizon, and thus a ‘coordinate artifact’. In fact, when expressed in terms of proper time of a freely falling observer, the observer will pass the event horizon at a finite proper time. In other words, just as Rindler coordinate can be smoothly extended into the whole of Minkowski space, so can Schwarzschild coordinates be extended into a black hole. There is one notable difference however. Namely, while the Minkowski observer can happily travel for ever after, a black hole observer will eventually crash into a space-time singularity in a finite proper time, and he/she can do nothing to prevent that, as was rigorously proven by this year’s Nobel Laureate Roger Penrose.

■ PROBLEM 18 Parallel transport on a sphere.

A vector V^μ is said to be parallel transported in the direction ν if its covariant derivative along that direction is zero,

$$\nabla_\nu V^\mu = \partial_\nu V^\mu + \Gamma_{\nu\sigma}^\mu V^\sigma = 0 . \quad (18.1)$$

Consider now a 2-dimensional sphere of unit radius,

$$ds^2 = d\theta^2 + \sin^2(\theta)d\phi^2 . \quad (18.2)$$

In this space vector V has just two components – V^θ and V^ϕ .

- (a) Compute the Christoffel symbols and show that the parallel transport equations along the ϕ -coordinate ($\theta = \text{const.}$) are

$$\partial_\phi V^\theta - \sin(\theta) \cos(\theta) V^\phi = 0 , \quad (18.3)$$

$$\partial_\phi V^\phi + \cot(\theta) V^\theta = 0 . \quad (18.4)$$

The two equations above can be seen as a system of two coupled ordinary differential equation of first order (with constant coefficients, since θ is fixed). We solve them by first constructing a second-order ordinary differential equation for one of the functions (in this case it can be done for both). Derive the following equation,

$$\partial_\phi^2 V^\theta = -\cos^2(\theta) V^\theta . \quad (18.5)$$

After solving this equation for V^θ , the solution for V^ϕ can be found from either (18.3) or (18.4).

- (b) Show that the full solution is

$$V^\theta = A \cos(\phi \cos \theta) + B \sin(\phi \cos \theta) , \quad (18.6)$$

$$V^\phi = -\frac{A \sin(\phi \cos \theta)}{\sin \theta} + \frac{B \cos(\phi \cos \theta)}{\sin \theta} , \quad (18.7)$$

where A and B are real constants (they determine the initial value of the vectors).

Next we want to understand how a vector is parallel transported along a triangle on a sphere. We are interested in the case where one of the sides of the triangle lies in the equator, subtending an angle α with the center of the sphere, and the other two sides connect the north pole to the equator (see Figure 1). The initial vector V at point X_1 is pointing in the ϕ -direction, namely $V^\phi = 1$ and $V^\theta = 0$. In the following, consider these three points on a sphere,

$$X_1 = (1, 0, 0) , \quad X_2 = (\cos \alpha, \sin \alpha, 0) , \quad X_3 = (0, 0, 1) . \quad (18.8)$$

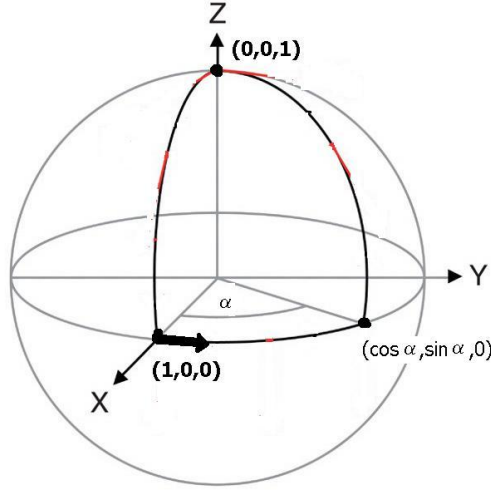


Figure 1: Triangle on a sphere

- (c) Verify that the vector V remains unchanged under parallel transport from X_1 to X_2 , i.e. that it remains in the ϕ direction. Make use of the solutions (18.6) and (18.7) you found.
- (d) Find how the vector is transported from X_2 to X_3 . In this case you can exploit the symmetries of the system: you can rotate your frame in such a way that the X_2X_3 section lies on the equator. In this frame the vector V has just the component V^θ .
- (e) Finally, using a similar trick, find how the vector changes if it is transported from X_3 back to X_1 . What happens if $\alpha = \frac{\pi}{2}$? Verify whether this result is compatible with what you expect from intuition (see Figure 1.)

For (d) and (e) it is enough to sketch and explain in words how the vector is parallel transported. Nevertheless, you are encouraged to construct explicitly all parallel transports required to solve this problem.

SOLUTION: (a) The non-vanishing Christoffel symbols for the sphere are,

$$\Gamma_{\phi\phi}^\theta = -\sin(\theta)\cos(\theta), \quad \Gamma_{\theta\theta}^\phi = \Gamma_{\phi\theta}^\phi = \cot(\theta). \quad (18.9)$$

The corresponding parallel transport equations are then,

$$\begin{aligned} \frac{dV^\theta}{d\tau} - \sin(\theta)\cos(\theta)\frac{d\phi}{d\tau}V^\phi &= 0 \\ \frac{dV^\phi}{d\tau} + \cot(\theta)\left[\frac{d\theta}{d\tau}V^\phi + \frac{d\phi}{d\tau}V^\theta\right] &= 0 \end{aligned} \quad (18.10)$$

For the problem at hand, $d\theta/d\tau = 0$, one can parametrize geodesics with ϕ , such that Eqs. (18.10) simplify to,

$$\begin{aligned} \frac{dV^\theta}{d\phi} - \sin(\theta)\cos(\theta)V^\phi &= 0 \\ \frac{dV^\phi}{d\phi} + \cot(\theta)V^\theta &= 0, \end{aligned} \quad (18.11)$$

which are the desired equations (18.3–18.4). These two equations can be easily decoupled, to produce two second order equations,

$$\begin{aligned} \frac{d^2V^\theta}{d\phi^2} + \cos^2(\theta)V^\theta &= 0 \\ \frac{d^2V^\phi}{d\phi^2} + \cos^2(\theta)V^\phi &= 0, \end{aligned} \quad (18.12)$$

where $\theta = \text{const.}$ should be considered as a parameter (labeling the curve).

(b) The general solutions to Eqs. (18.12) are easily constructed,

$$\begin{aligned} V^\theta &= A \cos(\phi \cos(\theta)) + B \sin(\phi \cos(\theta)) \\ V^\phi &= C \cos(\phi \cos(\theta)) + D \sin(\phi \cos(\theta)), \end{aligned} \quad (18.13)$$

Not all constants A, B, C, D can be chosen at will. Indeed, inserting (18.13) into the first order equations (18.12) gives the following two conditions,

$$C = \frac{B}{\sin(\theta)}, \quad D = -\frac{A}{\sin(\theta)}. \quad (18.14)$$

Upon inserting these into (18.13) gives the solution in the desired form (18.6–18.7).

(c) The path from $X_1 = (1, 0, 0)$ to $X_2 = (\cos(\alpha), \sin(\alpha), 0)$ is along a curve $\theta = \pi/2 = \text{const.}$, along which, $\cos(\theta) = 0$ and $\sin(\theta) = 1$, such that the parallel transport equations (18.6–18.7) apply. Along this equatorial trajectory and they simplify to,

$$V^\theta = A, \quad V^\phi = B. \quad (18.15)$$

Matching this with the initial condition $V^\theta = 0, V^\phi = 1$ yields $V^\theta(\phi) = 0$ and $V^\phi(\phi) = 1$, *i.e.* the components of V^μ do not change along that path, which is what was the task to prove.

(d) Here we ought to consider parallel transport from $X_2 = (\cos(\alpha), \sin(\alpha), 0)$ to $X_3 = (0, 0, 1)$. From figure 1 we see that along the path θ changes from $\theta = \pi/2$ at the equator (X_2) to $\theta = 0$ at the north pole (X_3), while ϕ remains constant. With this in mind, we see that for part (d) (and also for part (e)) we have $d\phi/d\tau = 0$, such that can consider ϕ to be a parameter independent of the position on the curve. With this in mind, the parallel transport equations (18.10) simplify to,

$$\begin{aligned} \frac{dV^\theta}{d\tau} &= 0 \\ \frac{dV^\phi}{d\tau} + \cot(\theta) \frac{d\theta}{d\tau} V^\phi &= 0. \end{aligned} \quad (18.16)$$

By considering $\tau = \tau(\theta)$, these equations simplify to,

$$\begin{aligned} \frac{dV^\theta}{d\theta} &= 0 \\ \frac{dV^\phi}{d\theta} + \cot(\theta) V^\phi &= 0. \end{aligned} \quad (18.17)$$

The first equation is easy to solve, $V^\theta = \text{const.}$ along the path. The second equation can be rewritten as, $d[\sin(\theta)V^\phi]/d\theta = 0$. This can be easily integrated out to give,

$$V^\phi(\theta) = \frac{E}{\sin(\theta)}, \quad (18.18)$$

where E is a constant to be determined by the boundary condition. The vector that we ought to parallel transport has the components $V^\theta = 0, V^\phi = 1$ at $\phi = \alpha, \theta = \pi/2$, at which $\sin(\theta) = 1$. When this is inserted into (18.18) this fixes the constant $E = 1$, such that we have along the curve connecting X_2 to X_3 ,

$$V^\theta = 0, \quad V^\phi = \frac{1}{\sin(\theta)}. \quad (18.19)$$

This means that V^ϕ blows up at the north pole (X_3), at which $\theta = 0$. This singularity is a coordinate artifact, as $V_\theta = \sin(\theta)$ vanishes at the north pole, such that the norm of the vector, $\|V^\mu\| = \sqrt{V_\mu V^\mu} = 1$ remains finite, and constant along the curve, which is what one would expect from a parallel transport of a scalar quantity. (A good quantity to consider is vector density, defined by, $\tilde{V}^\mu = \sqrt{|g|}V^\mu$, whose components remain constant along the above curve.)

(e) Transporting back from X_3 to X_2 , the same equations apply as in part (d), with the difference that now the path is along $\phi = 0$ from $\theta = 0$ to $\theta = \pi/2$. At the north pole one ought to be careful with the initial condition, since the end point the components are expressed in the frame at $\phi = \alpha$, while the parallel transport curve is along $\phi = 0$. That means that, before one starts parallel transport, one ought to rotate the initial vector from $\phi = \alpha$ to $\phi = 0$. To make this procedure clearer, this can be done along a curve that is infinitesimally away from the north pole at an angle $\delta\theta = \theta_0 \ll 1$. This parallel transport is what we already did in part (b), but now in the opposite sense. To solve the problem, first note that Eqs. (18.7) apply, and when adapted to this problem they can be written as,

$$\begin{aligned} V^\theta &= A \cos(\phi) + B \sin(\phi) \\ \sin(\theta_0)V^\phi &= B \cos(\phi) - A \sin(\phi), \end{aligned} \quad (18.20)$$

with the initial conditions at X_3 (at which $\phi = \alpha$ and $\theta = \theta_0$) are given by, $V^\phi = 0$ and $\sin(\theta_0)V^\theta = 1$. When these are inserted into (18.20) one obtains $B = \cos(\alpha)$, $A = -\sin(\alpha)$. Upon inserting these into (18.20) they simplify to,

$$\begin{aligned} V^\theta &= -\sin(\alpha) \cos(\phi) + \cos(\alpha) \sin(\phi) = \sin(\phi - \alpha) \\ \sin(\theta_0)V^\phi &= \cos(\alpha) \cos(\phi) + \sin(\alpha) \sin(\phi) = \cos(\phi - \alpha), \end{aligned} \quad (18.21)$$

This means that the initial components of the vector (to be parallel transported from X_3 back to X_1) are given in (18.21) with $\phi = 0$, $V^\theta = -\sin(\alpha)$, $\sin(\theta_0)V^\phi = \cos(\alpha)$. When parallel transported from X_3 to X_1 , along which V^θ and $\sin(\theta)V^\phi$ remain conserved, and we have at X_1 ,

$$V^\theta = -\sin(\alpha), \quad V^\phi = \cos(\alpha). \quad (18.22)$$

For $\alpha \neq 0$ these are different from the components one started with, $V^\theta = 0$ and $V^\phi = 1$. In fact, the components (18.22) can be obtained from the original components by applying the following rotation,

$$R(\alpha) = \begin{pmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{pmatrix} \quad (18.23)$$

on the original components, *i.e.* the vector has rotated by an angle α . The difference can be explained by the fact that the sphere is curved, and the rotation angle is proportional to the integrated curvature. The Gauss' curvature of the unit sphere is one, $\kappa = 1$, such that the integrated Gauss' curvature of the whole sphere is equal to its volume, $\int_{S^2} \kappa dV = 4\pi$. The volume of the segment of angle α is α (when $\alpha = 2\pi$ one obtains the hemisphere). Therefore we conclude that the vector has rotated by the amount equal to the Gauss' curvature of the segment around which the vector was parallel transported. While this result was simple to verify for the case of the triangular segment on the 2-sphere, it holds much more generally, *i.e.* vectors parallel transported around a closed curve will rotate by the amount given by the integrated curvature over the volume of which the curve is the boundary. Note that on curved manifolds such a volume is not uniquely given. Nevertheless, the rotation of a vector is unique, which means that this expresses a topological invariant of curvature. There is analogy with electromagnetism, in which if there is a magnetic flux through a surface, then the integral of the electromagnetic field along a closed curve around the flux is a topological invariant, and moreover it is quantized.